NetScope X1 main board, rev A draft: BLOCK-LEVEL schematic.

Generated from hardware/netlist/netscope-x1.yaml. Do not edit by hand; edit the YAML and run tools/gen_kicad.py.

Sheets marked REAL CAPTURE in their title block use library symbols and real pins (tools/main_sheets.py): the rest are block level:

Each part is a box with one pin per FUNCTION. Pin numbers are placeholders, bundles (RGMII, MDI, USB...) are one pin.

No footprints, no decoupling or other datasheet application parts. Not a fabrication schematic.

Domains: SYS = system side, LINE = cable conductors, POE = network-powered. Barrier parts have dashed outlines.

Checks: tools/check_kicad.py runs KiCad ERC and compares KiCad's exported netlist with the YAML netlist.

Port A: jack, relays, PoE magnetics, protection

File: 01_port_a.kicad_sch

Test-port MAC and PHY, uplink port

File: 02_test_port_mac_and_phy.kicad_sch

Wiremap front end

File: 03_wiremap_front_end.kicad_sch

PoE measurement island (isolated)

File: 04_poe_measurement_island_isolated.kicad_sch

Processor module, display, fiber port, options, debug

File: 05_processor_module.kicad_sch

Coprocessor, keys, indicators

File: 06_coprocessor.kicad_sch

Power: USB-C, charger, battery, rails

File: 07_power.kicad_sch

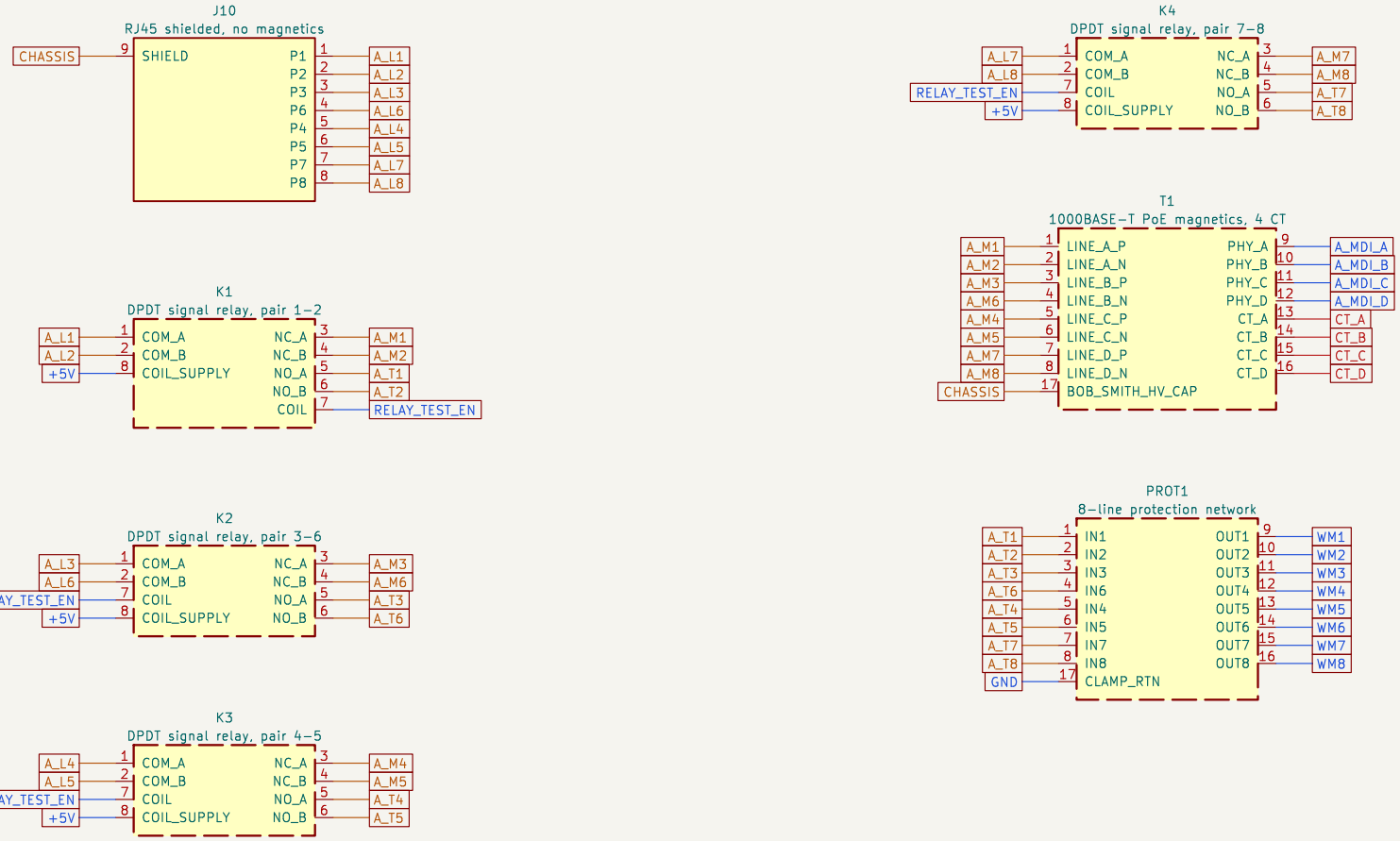
NetScope X1 main board — block schematic, rev A draft
Generated from hardware/netlist/netscope-x1.yaml by tools/gen_kicad.py
BLOCK LEVEL: function-named pins with placeholder numbers. Not for fabrication.
MT Tech Services LLC

Sheet: /
File: netscope-x1.kicad_sch

Title: NetScope X1 main board — sheet index

Size: A3	Date:	**Rev: A-draft**
KiCad E.D.A. 9.0.9		Id: 1/8

Block-level sheet. Dashed outline = isolation barrier part. Net names are global.



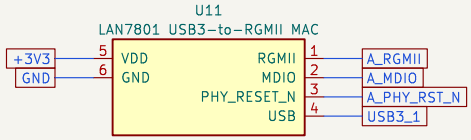
NetScope X1 main board — block schematic, rev A draft
Generated from hardware/netlist/netscope-x1.yaml by tools/gen_kicad.py
BLOCK LEVEL: function-named pins with placeholder numbers. Not for fabrication.
MT Tech Services LLC

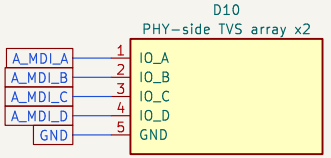
Sheet: /Port A: jack, relays, PoE magnetics, protection/
File: 01_port_a.kicad_sch

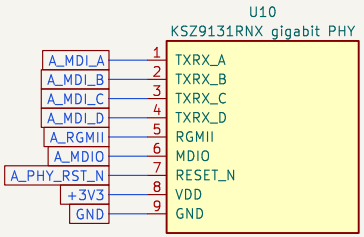
Title: Port A: jack, relays, PoE magnetics, protection

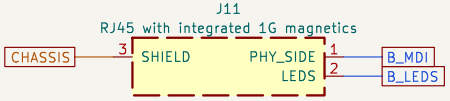
Size: A4	Date:	**Rev: A-draft**
KiCad E.D.A. 9.0.9		Id: 2/8

Block-level sheet. Dashed outline = isolation barrier part. Net names are global.









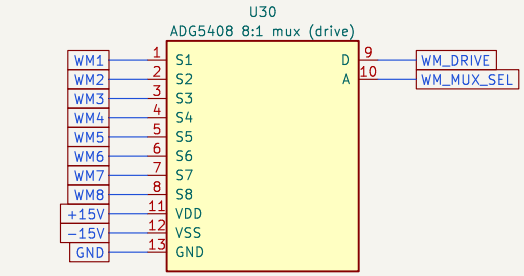
NetScope X1 main board — block schematic, rev A draft
Generated from hardware/netlist/netscope-x1.yaml by tools/gen_kicad.py
BLOCK LEVEL: function-named pins with placeholder numbers. Not for fabrication.
MT Tech Services LLC

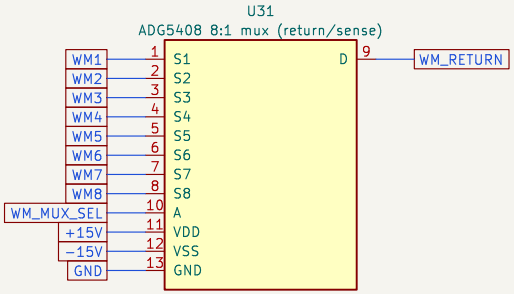
Sheet: /Test-port MAC and PHY, uplink port/
File: 02_test_port_mac_and_phy.kicad_sch

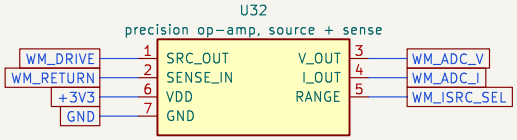
Title: Test-port MAC and PHY, uplink port

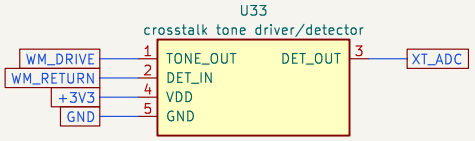
Size: A4	Date:	**Rev: A-draft**
KiCad E.D.A. 9.0.9		Id: 3/8

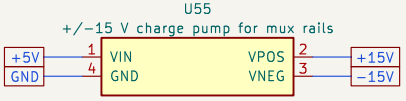
Block-level sheet. Dashed outline = isolation barrier part. Net names are global.











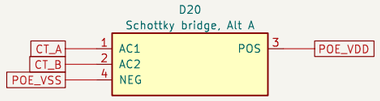
NetScope X1 main board — block schematic, rev A draft
Generated from hardware/netlist/netscope-x1.yaml by tools/gen_kicad.py
BLOCK LEVEL: function-named pins with placeholder numbers. Not for fabrication.
MT Tech Services LLC

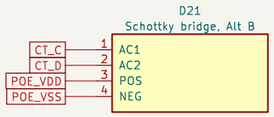
Sheet: /Wiremap front end/
File: 03_wiremap_front_end.kicad_sch

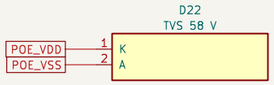
Title: Wiremap front end

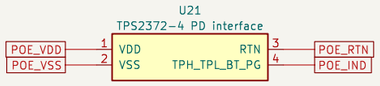
Size: A4	Date:	**Rev: A-draft**
KiCad E.D.A. 9.0.9		Id: 4/8

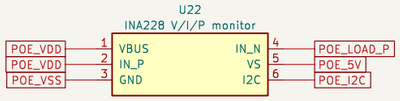
Block-level sheet. Dashed outline = isolation barrier part. Net names are global.



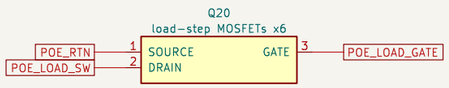


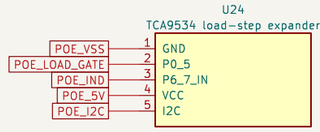


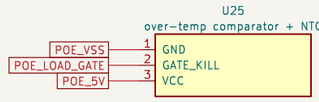


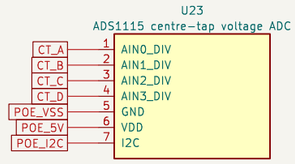


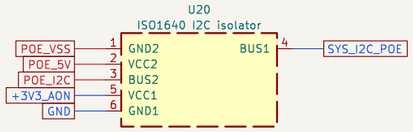


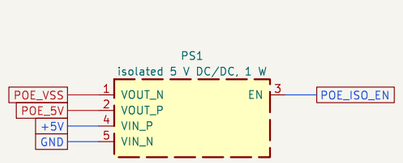












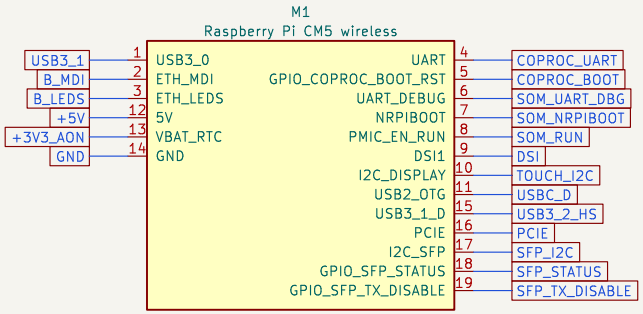
NetScope X1 main board — block schematic, rev A draft
Generated from hardware/netlist/netscope-x1.yaml by tools/gen_kicad.py
BLOCK LEVEL: function-named pins with placeholder numbers. Not for fabrication.
MT Tech Services LLC

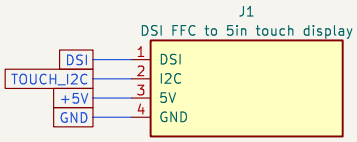
Sheet: /PoE measurement island (isolated)/
File: 04_poe_measurement_island_isolated.kicad_sch

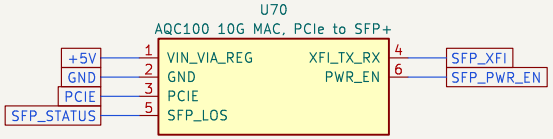
Title: PoE measurement island (isolated)

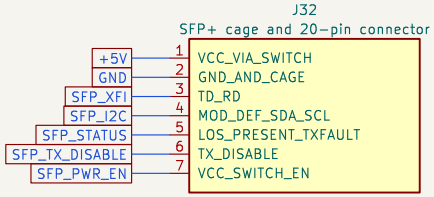
Size: A3	Date:	**Rev: A-draft**
KiCad E.D.A. 9.0.9		Id: 5/8

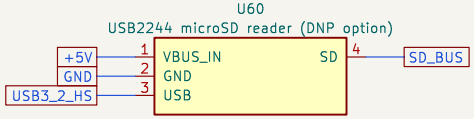
Block-level sheet. Dashed outline = isolation barrier part. Net names are global.

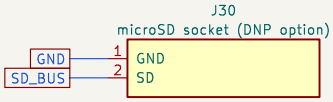


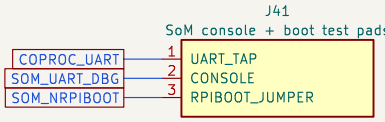












NetScope X1 main board — block schematic, rev A draft
Generated from hardware/netlist/netscope-x1.yaml by tools/gen_kicad.py
BLOCK LEVEL: function-named pins with placeholder numbers. Not for fabrication.
MT Tech Services LLC

Sheet: /Processor module, display, fiber port, options, debug/
File: 05_processor_module.kicad_sch

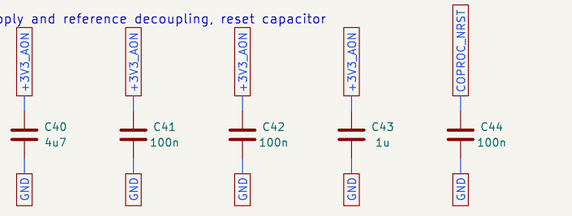
Title: Processor module, display, fiber port, options, debug

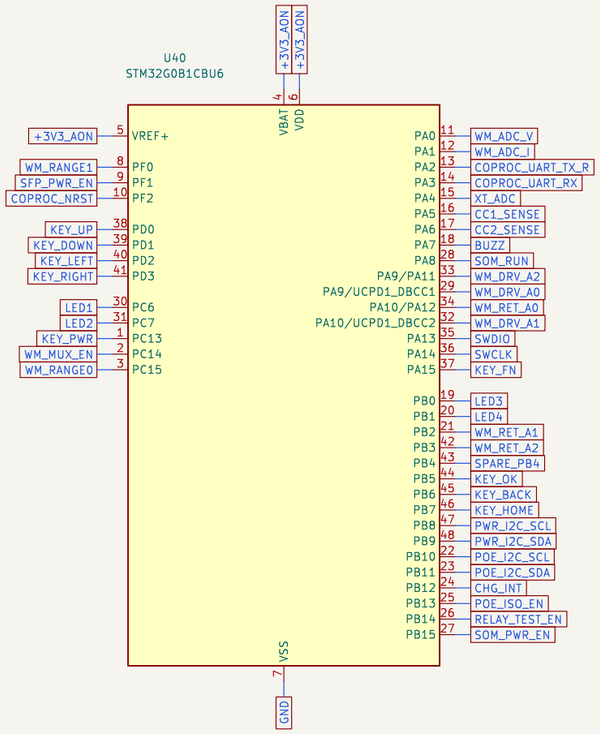
Size: A4	Date:	**Rev: A-draft**
KiCad E.D.A. 9.0.9		Id: 6/8

Coprocessor: STM32G0B1CBU6, always powered. Sheet captured for real: library symbols, real pin numbers.

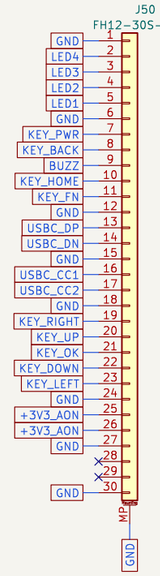
UNTESTED. Pin functions are checked against the library's alternate-function list by tools/check_kicad.py.

Supply and reference decoupling, reset capacitor

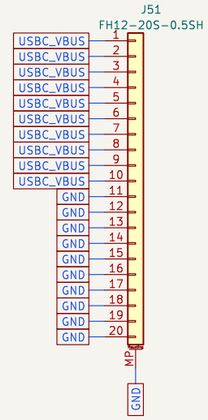




J50: 30-way signal cable to the lower board



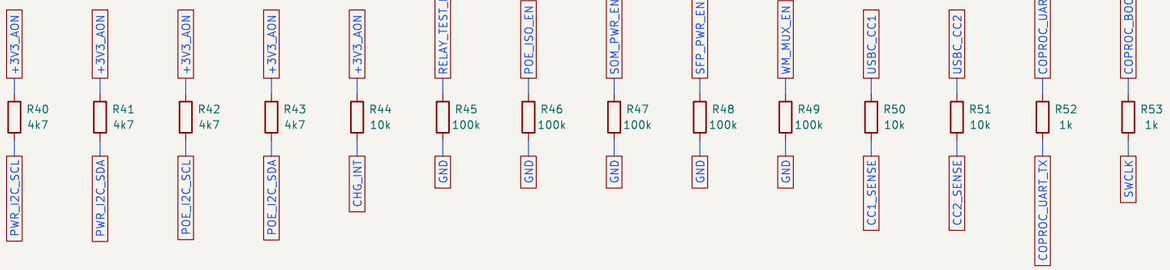
J51: 20-way power cable



Pin n here meets pin n on the lower board (cable with contacts on opposite faces).
That mapping is derived from the placement, not yet confirmed with parts in hand.

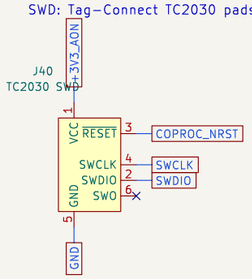
No crystal: internal 16 MHz oscillator. PA9/PA10 pins are UCPD dead-battery pins; used here as plain outputs. Boot entry for updates from the SoM (BOOT0 shares PA14) depends on option bits; to be verified.

Bus pull-ups, hold-off pull-downs (power and relay controls stay off while the MCU is in reset), series parts



R52 limits current into the SoM's pin while the SoM is unpowered; firmware must also float the line then.

SWD: Tag-Connect TC2030 pads, no part fitted





Generated by tools/gen_kicad.py from tools/main_sheets.py
REAL CAPTURE: library symbols and real pin numbers. UNTESTED.
MT Tech Services LLC

Sheet: /Coprocessor, keys, indicators/
File: 06_coprocessor.kicad_sch

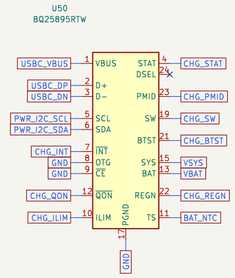
Title: Coprocessor, keys, indicators

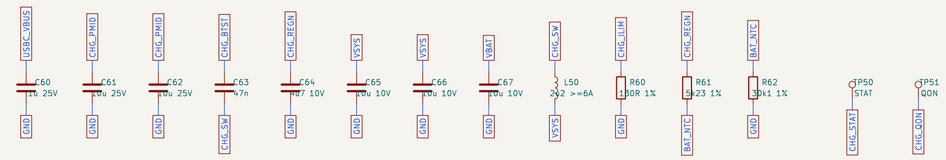
Size: A3	Date:	**Rev: A-draft**
KiCad E.D.A. 9.0.9		Id: 7/8

Power: charger and power path, fuel gauge, 5 V boost, two 3.3 V bucks. Sheet captured for real.

UNTESTED. Component values are data-sheet starting points, not a verified design: see hardware/kicad/README.md.

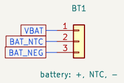
Charger and power path: BQ25895. USB-C input from the lower board (J51), system rail VSYS, battery VBAT.

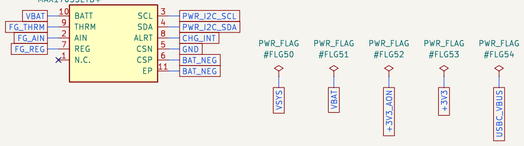


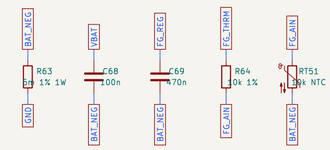


R60 caps the input current at 3.0 A or less (130R with the part's worst-case gain). Charging is enabled (CE low) and OTG boost is disabled (OTG low); current and voltage are set over I2C by the coprocessor.
R61/R62 are the data sheet's values for a 103AT thermistor: recalculate for the pack's actual NTC.
D+/D- sit on the USB data pair in parallel with the SoM, without the DSEL switch TI describes; to be bench-tested.

Battery connection and fuel gauge: MAX17055. Pack negative reaches ground through the 5 mOhm sense resistor.

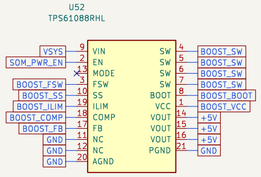


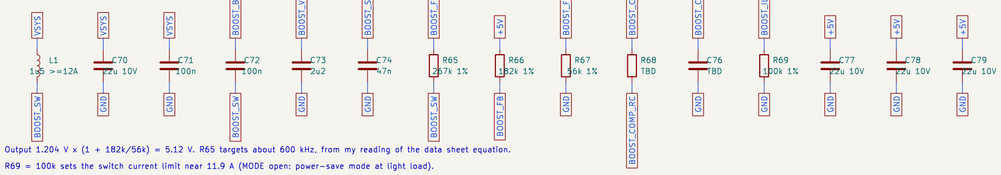




5 mOhm, not the data sheet's 10: the boost can draw over 7 A from the pack and 10 mOhm measures only to 5.1 A.
RT51 is a board thermistor for the gauge; the pack's own NTC goes to the charger only (two bias networks cannot share one thermistor). U51 and BT1 have no footprint yet: see the README.

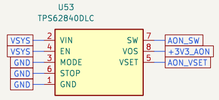
5.1 V boost: TPS61088, enabled by the coprocessor (SOM_PWR_EN). Feeds the SoM and everything on +5V.

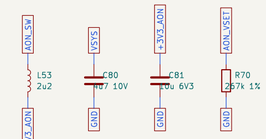




Output 1.204 V x (1 + 182k/56k) = 5.12 V. R65 targets about 600 kHz, from my reading of the data sheet equation.
R69 = 100k sets the switch current limit near 11.9 A (MODE open: power-save mode at light load).
R68 / C76 (loop compensation) are NOT chosen: the data sheet has no 5 V example. The board must not be built with them open.

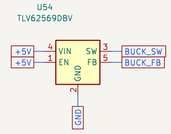
Always-on 3.3 V: TPS62840 from VSYS (coprocessor, key pull-ups)

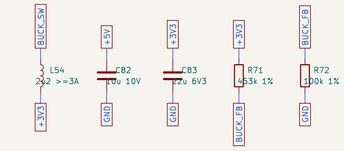




R70 = 267k selects 3.3 V. Always enabled (EN to VSYS).

3.3 V for the Ethernet and analogue parts: TLV62569 from +5V





0.6 V x (1 + 453k/100k) = 3.32 V. Follows +5V (EN tied to the input).

Generated by tools/gen_kicad.py from tools/main_sheets.py
REAL CAPTURE: library symbols and real pin numbers. UNTESTED.
MT Tech Services LLC

Sheet: /Power: USB-C, charger, battery, rails/
File: 07_power.kicad_sch

Title: Power: USB-C, charger, battery, rails

Size: A2	Date:	**Rev: A-draft**
KiCad E.D.A. 9.0.9		Id: 8/8